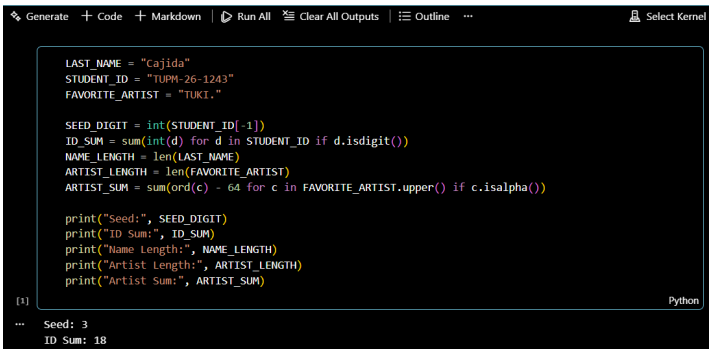
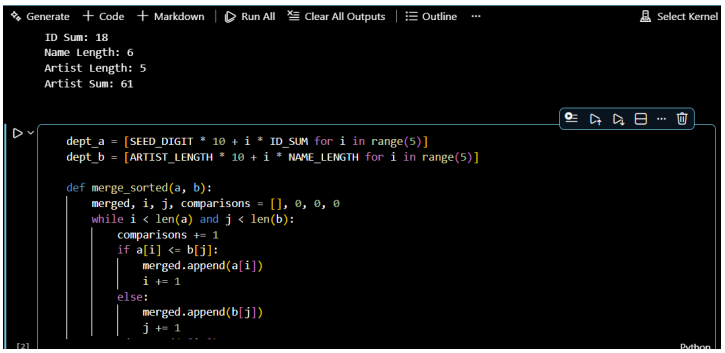
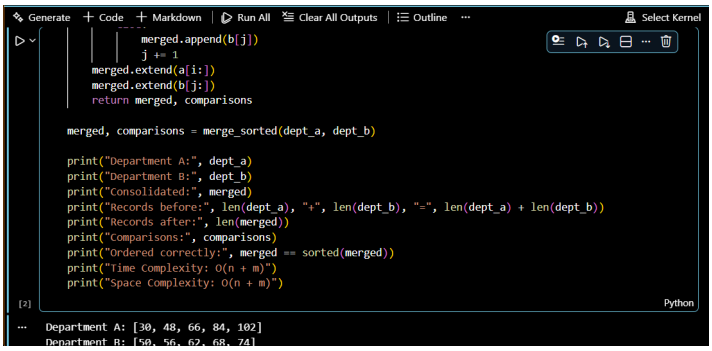
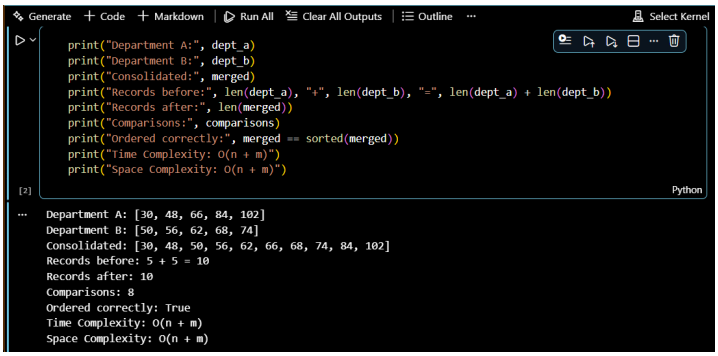

Generate
Code
Markdown
Run All
Clear All Outputs
Outline
Select Kernel
LAST_NAME = "Cajida"
STUDENT_ID = "TUPM-26-1243"
FAVORITE_ARTIST = "TUKI."
SEED_DIGIT = int(STUDENT_ID[-1])
ID_SUM = sum(int(d) for d in STUDENT_ID if d.isdigit())
NAME_LENGTH = len(LAST_NAME)
ARTIST_LENGTH = len(FAVORITE_ARTIST)
ARTIST_SUM = sum(ord(c) - 64 for c in FAVORITE_ARTIST.upper() if c.isalpha())
print("Seed:", SEED_DIGIT)
print("ID Sum:", ID_SUM)
print("Name Length:", NAME_LENGTH)
print("Artist Length:", ARTIST_LENGTH)
print("Artist Sum:", ARTIST_SUM)
[1]
Python
Seed: 3
ID Sum: 18


Generate
Code
Markdown
Run All
Clear All Outputs
Outline
Select Kernel
ID Sum: 18
Name Length: 6
Artist Length: 5
Artist Sum: 61
dept_a = [SEED_DIGIT * 10 + i * ID_SUM for i in range(5)]
dept_b = [ARTIST_LENGTH * 10 + i * NAME_LENGTH for i in range(5)]
def merge_sorted(a, b):
merged, i, j, comparisons = [], 0, 0, 0
while i < len(a) and j < len(b):
comparisons += 1
if a[i] <= b[j]:
merged.append(a[i])
i += 1
else:
merged.append(b[j])
j += 1
[2]
Python


Generate
Code
Markdown
Run All
Clear All Outputs
Outline
Select Kernel
merged.append(b[j])
j += 1
merged.extend(a[i:])
merged.extend(b[j:])
return merged, comparisons
merged, comparisons = merge_sorted(dept_a, dept_b)
print("Department A:", dept_a)
print("Department B:", dept_b)
print("Consolidated:", merged)
print("Records before:", len(dept_a), "+", len(dept_b), "=", len(dept_a) + len(dept_b))
print("Records after:", len(merged))
print("Comparisons:", comparisons)
print("Ordered correctly:", merged == sorted(merged))
print("Time Complexity: O(n + m)")
print("Space Complexity: O(n + m)")
[2]
Python
Department A: [30, 48, 66, 84, 102]
Department B: [50, 56, 62, 68, 74]


Generate
Code
Markdown
Run All
Clear All Outputs
Outline
Select Kernel
print("Department A:", dept_a)
print("Department B:", dept_b)
print("Consolidated:", merged)
print("Records before:", len(dept_a), "+", len(dept_b), "=", len(dept_a) + len(dept_b))
print("Records after:", len(merged))
print("Comparisons:", comparisons)
print("Ordered correctly:", merged == sorted(merged))
print("Time Complexity: O(n + m)")
print("Space Complexity: O(n + m)")
[2]
Python
Department A: [30, 48, 66, 84, 102]
Department B: [50, 56, 62, 68, 74]
Consolidated: [30, 48, 50, 56, 62, 66, 68, 74, 84, 102]
Records before: 5 + 5 = 10
Records after: 10
Comparisons: 8
Ordered correctly: True
Time Complexity: O(n + m)
Space Complexity: O(n + m)